

第三章 刚体的定轴转动 (大题)

一. 刚体的运动 (平动+转动)

1. Def: 刚体: 可忽略形状与体积的改变

平动: 质心运动讨论 (各质点运动轨迹完全相同)

转动: 对点, 对轴 <定轴转动> 绕同一直线做圆周运动

定轴转动: 质元均做圆周运动. 圆心在一条直线上. 线量不同. 角量相同

质心: 质点系质量中心

质心的位矢: $\vec{r}_c = \frac{\sum m_i \vec{r}_i}{\sum m_i} = \frac{\sum m_i \vec{r}_i}{M}$ (对于某质点系, 质心位置固定)

分量式: $x_c = \frac{\sum m_i x_i}{M}$, $y_c = \frac{\sum m_i y_i}{M}$, $z_c = \frac{\sum m_i z_i}{M}$

质量连续分布 $x_c = \int x dm / M$, $y_c = \int y \frac{dm}{M}$, $z_c = \int z \frac{dm}{M}$

质心坐标系中质心位矢(0,0,0) (2个质点组成的质点系取质心处为 $x_c = 0$)

2. 刚体的定轴转动

(1) 函数 $\theta = \theta(t)$... 运动学方程, 逆时针方向为正

Δt 内角坐标增量 $\Delta\theta$ 为该时间间隔内的角位移. (rad)

用

角位移: $\Delta\theta$

(2) 角速度 $\omega = \frac{d\theta}{dt}$ (rad/s)

> 方向相同. 加速转动

角速度 $\omega = \frac{d\theta}{dt}$

角加速度 $\beta = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$

反之减速 rad/s²

角加速度 $\beta = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$

匀加速转动: $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\beta t^2$

$d\theta > 0 \Rightarrow \omega > 0$

(β 为常数) $\omega = \omega_0 + \beta t$

$\omega^2 = \omega_0^2 + 2\beta\theta$

△定轴转动特点

(3) 角量与线量

$s = r\theta$

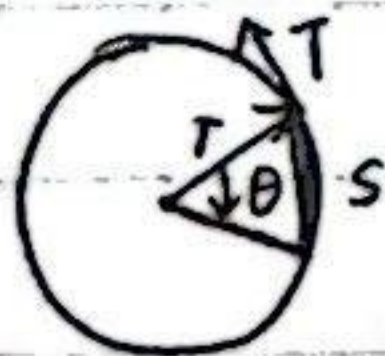
$\Rightarrow \frac{ds}{dt} = \frac{d\theta}{dt} r \Rightarrow v_t = r\omega$

切向 $a_t = \frac{dv_t}{dt} = r \frac{d\omega}{dt} = r\beta$ 角量

(向心)法向 $a_n = \frac{v_t^2}{r} = \frac{(r\omega)^2}{r} = \omega^2 r = v \cdot \omega = \frac{v^2}{r}$

线量

角量



① 圆周运动

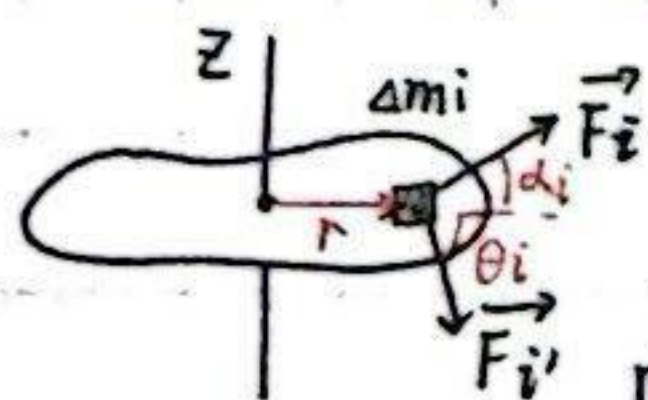
② 角坐标 θ

③ $\Delta\theta$ 大小一致

ω 一样

β 一样

***转动定律推导:



$$\vec{F} = \vec{F}_i + \vec{F}_{i'} = \Delta m_i \vec{a}_i$$

对转动有影响, 只研究切线方向 $\vec{F}_{i\tau} + \vec{F}_{i'\tau} = \Delta m_i \vec{a}_{i\tau} = \Delta m_i r \beta$

$$\text{同} \times \vec{r}: r \cdot F_i \sin \alpha_i + r F_{i'} \sin \theta_i = \Delta m_i r^2 \beta$$

$$\text{求和} \quad \sum r F_i \sin \alpha_i + \sum r F_{i'} \sin \theta_i = \boxed{\sum \Delta m_i r_i^2} \beta$$

合外力矩 合内力矩 $\Rightarrow 0$

$$\Rightarrow \sum r F_i \sin \alpha_i = \sum \Delta m_i r_i^2 \beta$$

$$\Rightarrow \vec{\tau} = J \vec{\beta} \quad (\vec{M} = J \vec{\beta})$$

合外力矩 转动惯量 $\times$ 角加速度

补充文字说明: 刚体定轴转动的角加速度与它所受的合外力矩成正比

与刚体的转动惯量成反比

$$\Rightarrow \beta \propto \frac{\vec{\tau}}{J}$$

$$\Rightarrow \vec{\tau} = J \beta = J \frac{d\omega}{dt}$$

$\Rightarrow \vec{\tau}$ 为 0 则 β 为 0, J 不可能为 0

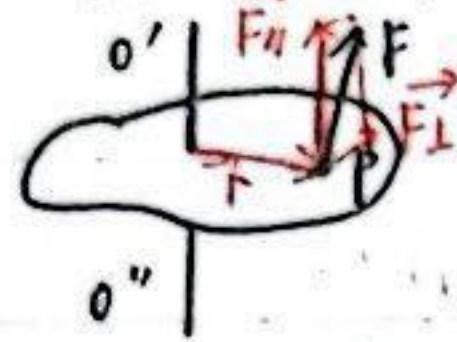
$\hookrightarrow \omega$ 不变为常数

二、刚体定轴转动定律

1. 力矩: Def: $\vec{M} = \vec{r} \times \vec{F}$

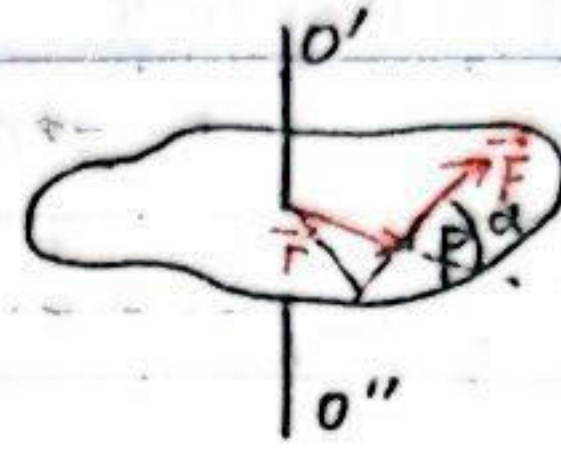
① $\vec{F}$ 在垂直 $O'O''$ 平面内 $\vec{M} = rF \sin \alpha$

② $\vec{F}$ 不在平面内



$$\vec{F} = \vec{F}_\perp + \vec{F}_\parallel$$

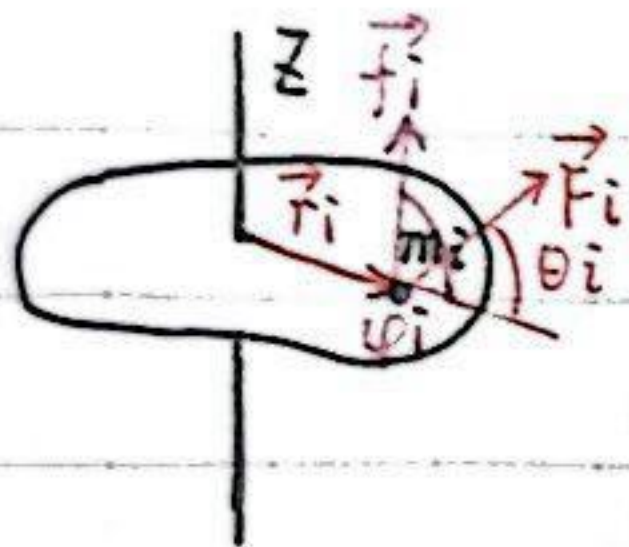
(对转动无贡献)



计算 $\vec{M}$ 时, 只考虑 $\vec{F}_\perp$ 的力矩 $\vec{M} = \vec{r} \times \vec{F}_\perp = rF_\perp \sin \theta$

2. 定律推导:

设某刚体绕轴转动, 取质量元 m_i , 到轴距离为 r_i



牛顿定律: $\vec{F}_i + \vec{f}_i = m_i \vec{a}_i$

加速度不同, 角加速度相同: $F_i \sin \theta_i + f_i \sin \phi_i = m_i a_i r$

同乘 r_i : $r_i F_i \sin \theta_i + r_i f_i \sin \phi_i = r_i m_i a_i r$

$\therefore a_i = r\beta \Rightarrow r_i F_i \sin \theta_i + r_i f_i \sin \phi_i = m_i r_i^2 \beta$

所有质元相加 $\sum r_i F_i \sin \theta_i + \sum r_i f_i \sin \phi_i = \sum m_i r_i^2 \beta$

$\sum \vec{r}_i \times \vec{F}_i = \tau_{\text{合外}}$ (合外力矩)

$\Rightarrow \sum r_i F_i \sin \theta_i = (\sum m_i r_i^2) \beta$

$\Rightarrow \vec{M} = J \vec{\beta}$ (可不同轴但平行)
同轴

3. 定轴转动定律: $\vec{M} = J \vec{\beta}$ (形如牛二: $\vec{F} = m \vec{a}$) m 反映平动惯性

外力对此轴线合力矩 = 转动惯量与角加速度的乘积

J 反映刚体的转动惯性, 不可能为 0 (转动惯性的量度)

$J = \sum m_i r_i^2$ (各质元质量与其到转轴垂直距离平方的乘积和)

$J = \int r^2 dm$ (质量均匀分布)

三、转动惯量的计算

相关因素 ① 转轴位置 ② 质量分布

线分布



$$dm = \lambda dl$$

(λ : 线密度)

面分布



$$dm = \sigma ds$$

(σ : 面密度)

体分布



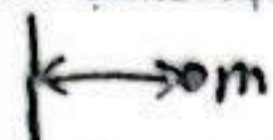
$$dm = \rho dV$$

(ρ : 体密度)

质心处 $J_{\min}$ (平行轴定理: $J = J_c + md^2$)

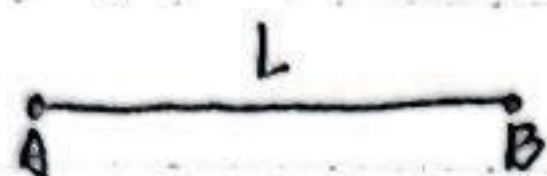
常见均匀刚体的转动惯量

顶点



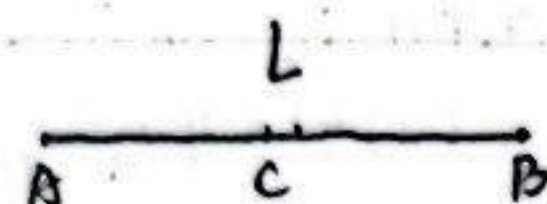
$$J = mr^2$$

细杆 (端点轴)



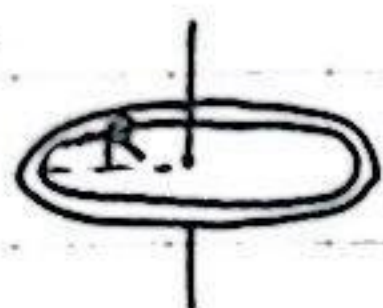
$$J = \frac{1}{3} mL^2$$

细杆 (质心轴)



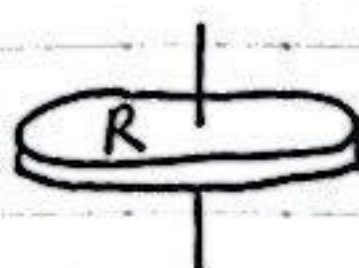
$$J = \frac{1}{12} mL^2$$

圆环 (圆筒) (环心)



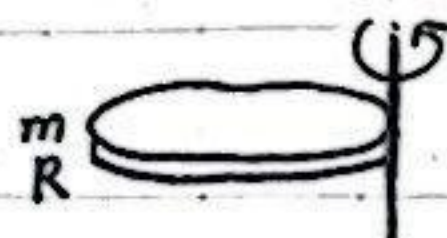
$$J = mR^2$$

圆盘 (圆柱) (柱心)



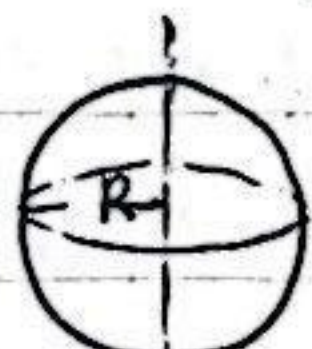
$$J = \frac{1}{2} mR^2$$

圆盘 (圆柱)



$$J = \frac{1}{2} mR^2 + mR^2$$

球壳

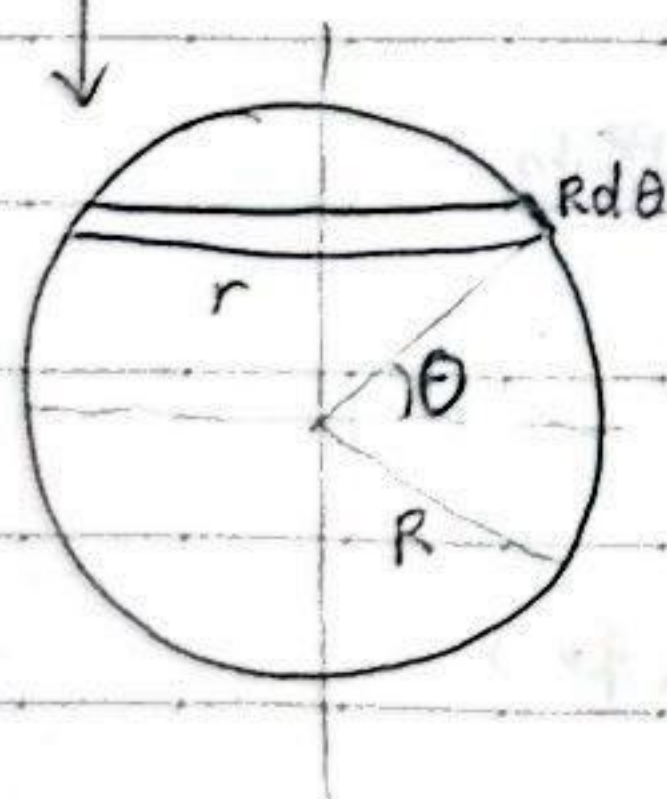


$$J = \frac{2}{3} mR^2$$

球体



$$J = \frac{2}{5} mR^2$$



$$dm = \sigma ds = \sigma 2\pi r \cdot R d\theta$$

$$dJ = r^2 dm = \sigma 2\pi r^3 R d\theta$$

$$J = \int_0^\pi 2\sigma 2\pi r^3 R d\theta$$

平行轴定理

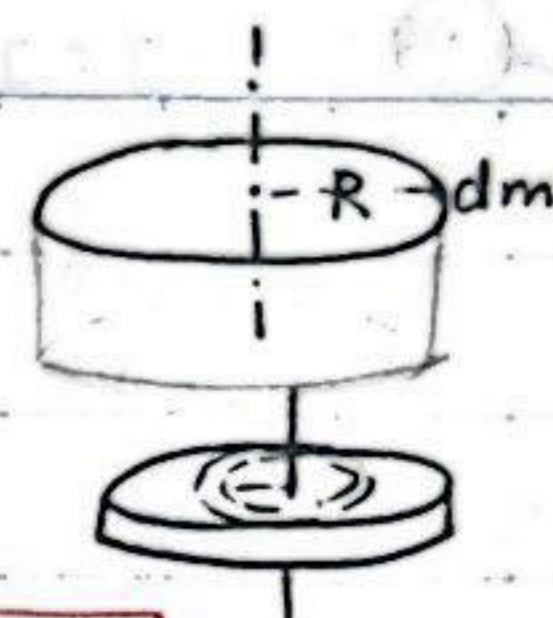
$$J_A = J_c + md^2$$

垂直轴定理

$$J_z = J_x + J_y$$

常见转动惯量

(1) 圆环 (圆桶): $J = \int R^2 dm = R^2 \int dm = \boxed{mR^2}$



(2) 圆盘 (半径 R , 厚度 L) (圆柱)

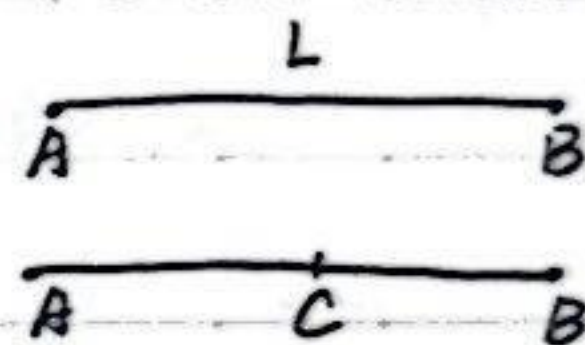
$\rho = \frac{m}{\pi R^2 L}$, 圆环: $dm = \rho \cdot 2\pi r L dr$, $dJ = dm \cdot r^2 = 2\pi L r^3 dr$

$\therefore J = \int_0^R dJ = \int_0^R \rho \cdot 2\pi L r^3 dr = \frac{1}{2} \rho \pi R^4 L \Rightarrow \boxed{J = \frac{1}{2} m R^2}$ (与厚度无关)

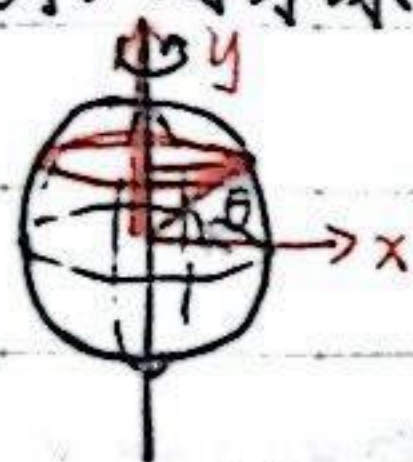
(3) 均匀细杆 a. 端点轴 $dm = \lambda dx$, $J_A = \int_0^L x^2 dm = \int_0^L \lambda x^2 dx = \boxed{\frac{mL^2}{3}}$

b. 质心轴

$J_C = \int_{-\frac{L}{2}}^{\frac{L}{2}} \lambda x^2 dx = \boxed{\frac{1}{12} mL^2}$



例: 均匀球体相对直径的转动惯量 $J = \frac{2}{5} MR^2$



圆盘累加 $\frac{1}{2} m$ 厚度为 dy , 高度为 y

$\sin \theta = \frac{y}{R}$, $r = R \cos \theta = \sqrt{R^2 - y^2}$, $dm = \rho dV = \rho \cdot \pi r^2 dy = \rho \pi (R^2 - y^2) dy$

$dJ = \frac{1}{2} dm \cdot r^2 = \frac{1}{2} \rho \pi (R^2 - y^2)^2 dy$

$\rho = \frac{4\pi R^3}{3M}$

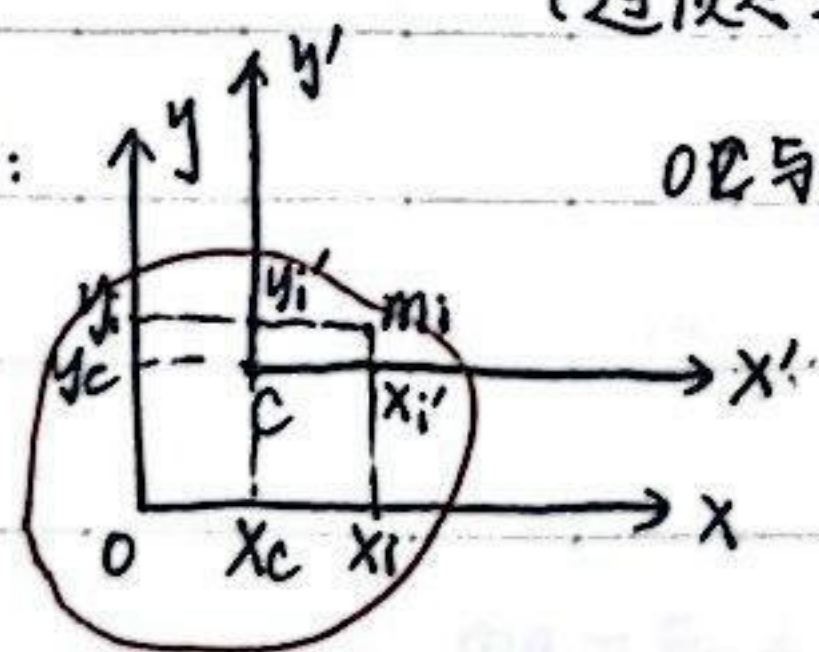
$J = \int_{-R}^R \frac{\rho}{2} \pi r^4 dy = \frac{2}{5} MR^2$

★ 平行轴定理: 任一轴与质心平行, 相距为 d , 刚体对其转动惯量为 J , 则有

$J_A = J_C + md^2$

(过质心轴的转动惯量)

证明:



OE 与 C 与纸面垂直, 带'为质心坐标系

$J = \sum m_i (x_i^2 + y_i^2)$

$= \sum m_i [(x_i' + x_c)^2 + (y_i' + y_c)^2]$

$= \sum m_i (x_i'^2 + y_i'^2) + 2x_c \sum m_i x_i' + 2y_c \sum m_i y_i' + (x_c^2 + y_c^2) \sum m_i$

$= \sum m_i (x_i'^2 + y_i'^2)$

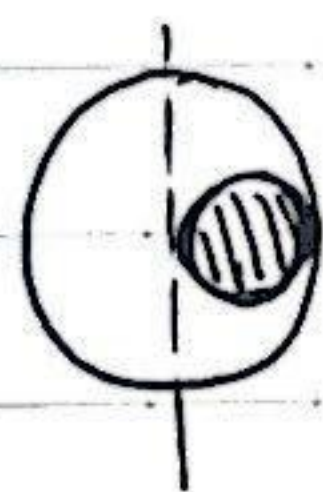
$Mx_c' = 0$

$My_c' = 0$

$+ (x_c^2 + y_c^2) \sum m_i$

为0

$\Rightarrow J = J_C + md^2$



★ 补偿法

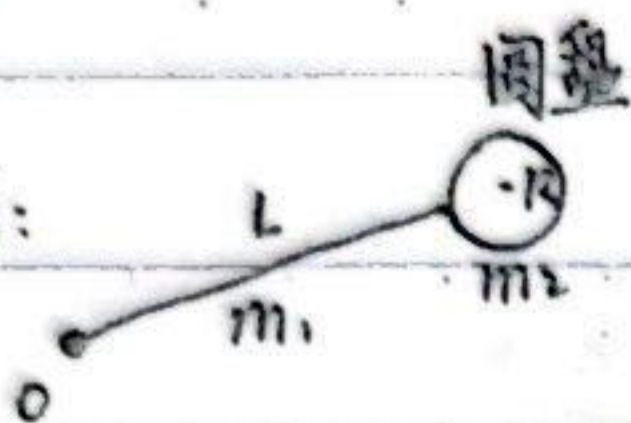
有空洞圆球: $J = J_1 - J_2$



相对质心

相对平行质心的轴

例:



$$J = J_1 + J_2$$

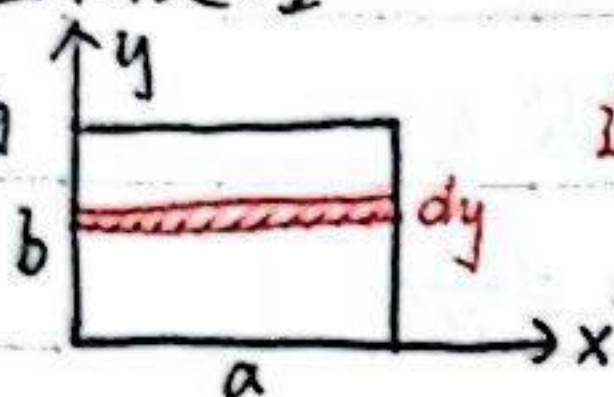
$$= \frac{1}{3} m_1 L^2 + (J_c + m_2 d^2)$$

$$= \frac{1}{3} m_1 L^2 + \frac{1}{2} m_2 R^2 + m_2 (R+L)^2$$

例: 一均匀质量长方形薄片长为 a , 宽为 b , 质量为 M . 求其转动惯量

解: 利用垂直轴定理

对 x 轴



选择分割成长度为 x , 宽度为 dy 的质点

$$dJ_x = dm r^2 = \sigma a dy \cdot y^2$$

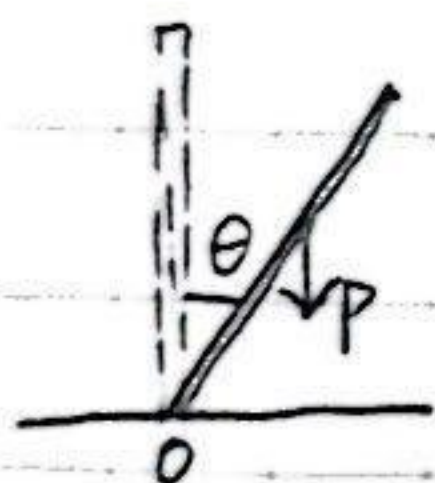
$$J_x = \int_0^a \sigma a y^2 dy = \frac{1}{3} \sigma a b^3$$

同理对 y 轴: $J_y = \frac{1}{3} \sigma a^3 b$

($\sigma(a^2 + b^2) = M$)

$$\therefore J_z = J_x + J_y = \frac{1}{3} \sigma a b (a^2 + b^2) = \frac{1}{3} M (a^2 + b^2)$$

例: 长为 L , 匀质细杆竖直放置, 受扰动后在重力作用下绕 O 转动, 计算与铅垂线呈 θ 角时 β 和 w



解: 只有重力有力矩 $Mg \times \bar{r} = J \cdot \beta$

$$Mg \cdot \frac{1}{2} L \sin \theta = \frac{1}{3} M L^2 \cdot \beta \Rightarrow \beta = \frac{3g \sin \theta}{2L}$$

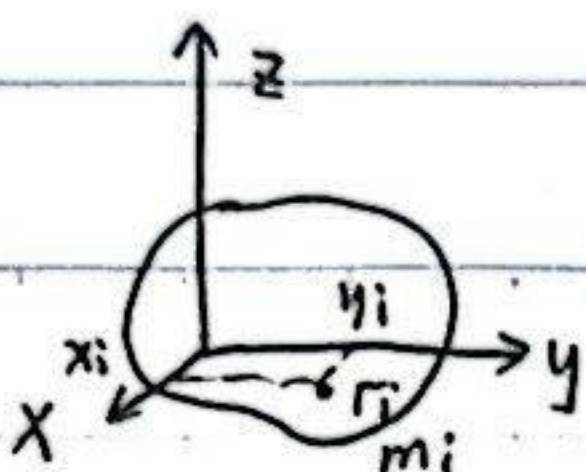
$$\frac{dw}{dt} = \beta$$

$$\int_0^w w dw = \int_0^\theta \frac{3g}{2L} \sin \theta d\theta$$

$$w = \sqrt{\frac{3g}{L} (1 - \cos \theta)}$$

△ 垂直轴定理: $J_z = J_x + J_y$

(无穷小厚度薄片)

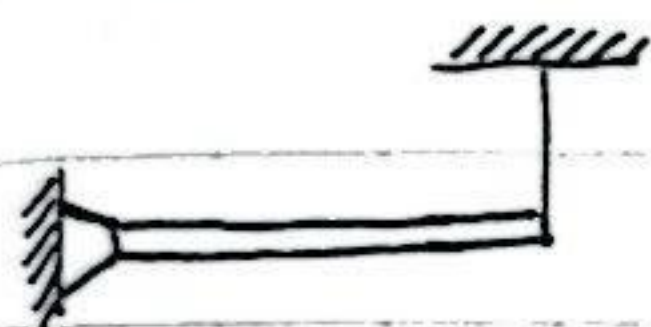


$$J_z = \sum m_i r_i^2 = \sum m_i (x_i^2 + y_i^2)$$

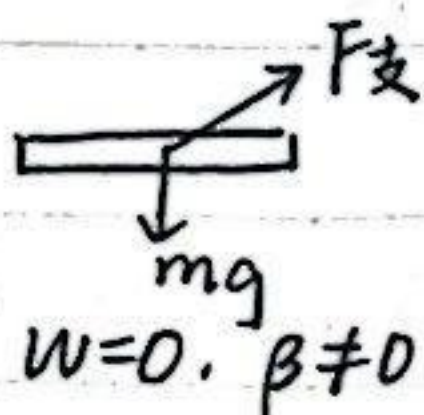
$$= \sum m_i x_i^2 + \sum m_i y_i^2 = J_x + J_y$$

四. 转动定律的应用

例1: 质量为 M , 长为 L 的匀质细杆水平放置. 一端铰链. 另一端用绳悬挂. 求剪断瞬时. 杆的角加速度及铰链的支撑力



解: 对杆



$$\vec{F} + m\vec{g} = m\vec{a}$$

$$\omega = 0, \beta \neq 0$$

杆绕一端转动, 只有重力有力矩



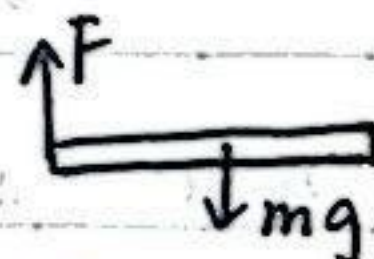
$$\vec{M} = mg \times d = \frac{1}{2}mgL$$

定轴转动定律: $\vec{M} = J\beta$

$$J = \frac{1}{3}mL^2$$

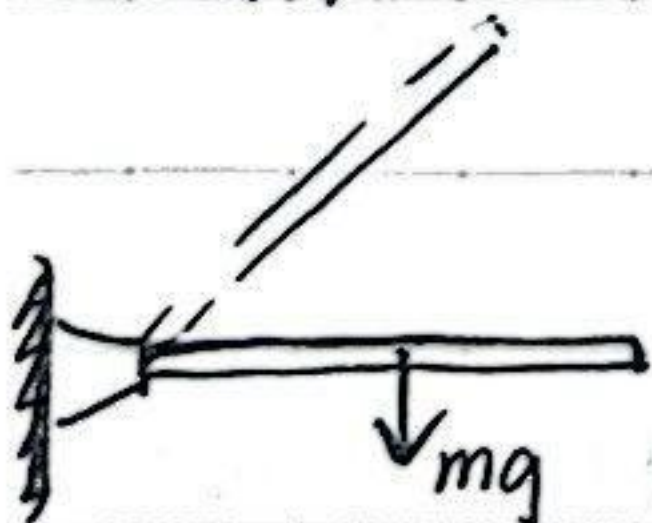
$$\text{解得 } \beta = \frac{3g}{2L}$$

⇒ 受力分析



$$\text{质心平动: } mg - F = ma_z = m \cdot \frac{1}{2}L\beta \Rightarrow F = m(g - a_z) = \frac{1}{4}mg$$

变例: 若杆从上方转到水平位置时 $\omega = \omega_0$. 此时 $\beta, F = ?$



解: 水平位置分析. 力矩仍为 $\vec{M} = \frac{1}{2}mgL$

$$\beta = \frac{3g}{2L}$$

$$\text{质心切向加速度 } a_t = r\beta = \frac{3g}{4}$$

$$\text{法向加速度 } a_n = \frac{v^2}{r} = \frac{(\omega r)^2}{r} = \omega^2 r = \frac{1}{2}\omega_0^2 L$$

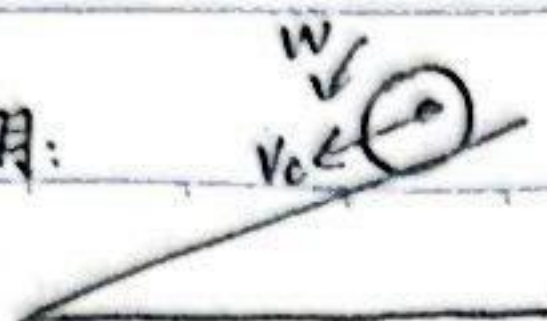
$$mg - F_z = ma_z \Rightarrow F_z = \frac{1}{4}mg$$

$$F_n = m\omega^2 r \Rightarrow F_n = \frac{1}{2}m\omega_0^2 L$$

$$\therefore F_{\text{合}} = \sqrt{(\frac{1}{4}mg)^2 + (\frac{1}{2}m\omega_0^2 L)^2}$$

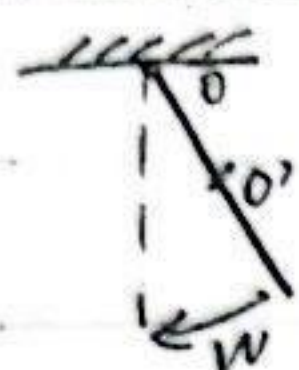
$$\tan \alpha = \frac{F_t}{F_n} = \frac{g}{2L\omega_0^2}$$

说明:



$$E_k = \frac{1}{2} m v_c^2 + \frac{1}{2} J \omega^2$$

质量集中

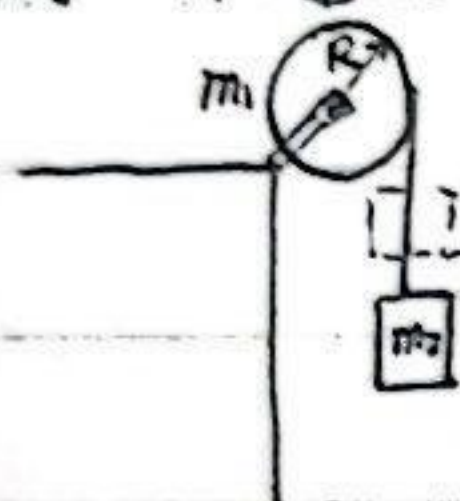


$$\text{绕 } O \text{ 点 } E_k = \frac{1}{2} J_O \omega^2$$

$$\text{绕 } O' \text{ 点 } E_k = \frac{1}{2} m v_{O'}^2 + \frac{1}{2} J_{O'} \omega^2$$

相等

★例. 求重锤从静止释放带动柱体旋转. 下降 h 时速率 v



解: 法一. 刚体转动定律 + 牛二解 a 得 v

法二. 动能定理

a. 单独分析: 对 m_2 : 拉力为 T (定值) $m_2 g h - T h = \frac{1}{2} m_2 v^2 - 0$ ①

对 m_1 : 转过角度 θ

$$A = \int_0^\theta \tau d\theta = T R \theta$$

只有拉力力矩 $\vec{\tau} = T \cdot R$

$$A = \frac{1}{2} J \omega_2^2 - \frac{1}{2} J \omega_1^2 = \frac{1}{2} m_1 R^2 \omega^2$$

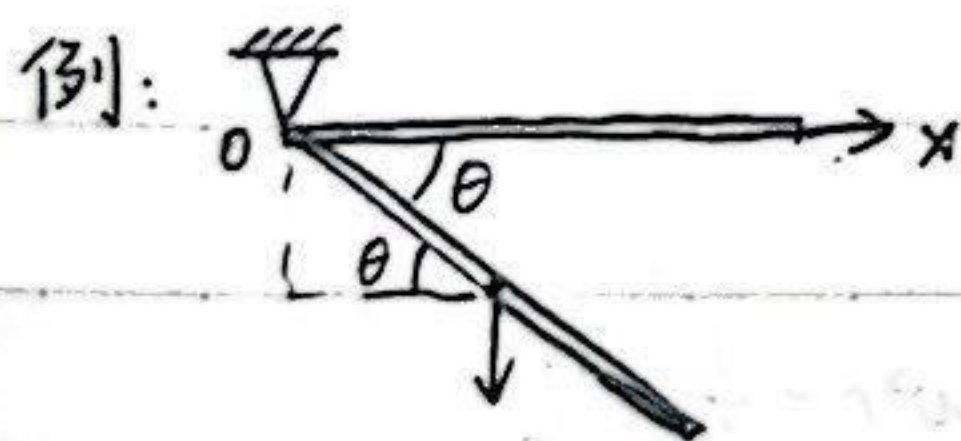
且有 $R \theta = h, v = R \omega$ ③

$$T R \theta = \frac{1}{2} m_1 R^2 \omega^2$$
 ②

合外力做功 内力做功 联立求解

b. 整体法: $m_2 g h + 0 = \frac{1}{2} m_2 v^2 + \frac{1}{2} J \omega^2$ 且 $v = R \omega \Rightarrow v = 2 \sqrt{\frac{m_2 g h}{m_1 + 2 m_2}}$

拉力等值反号

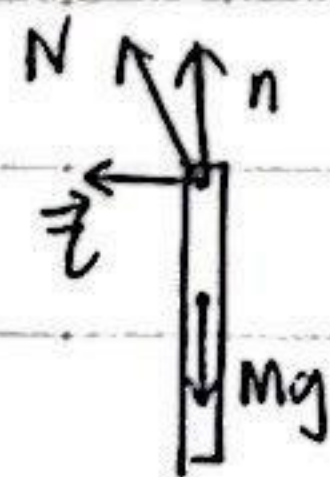


例: (1) 求 θ 角时 β 和 ω (2) 杆在铅垂位置时对支点的作用力

(1) 只有重力矩做功 机械能守恒 $\begin{cases} \frac{1}{2} L \sin \theta \cdot m g = \frac{1}{2} J \omega^2 \\ J = \frac{1}{3} m L^2 \end{cases} \Rightarrow \omega = \sqrt{\frac{3 g \sin \theta}{L}}$

$$\beta = \frac{d\omega}{dt} = \omega \frac{d\omega}{d\theta} = \frac{3 g \cos \theta}{2 L}$$

(2) 作用力时刻改变 $\Rightarrow$ 牛二分析. 自然坐标系



$$\vec{N} + \vec{G} = m \vec{a}_c$$

$$N_n - m g = m \frac{v^2}{R}$$

$$R = \frac{1}{2} L, v = \frac{1}{2} \omega L = \frac{1}{2} \sqrt{3 g L}$$

$$N_z = m a_{cz} = 0$$

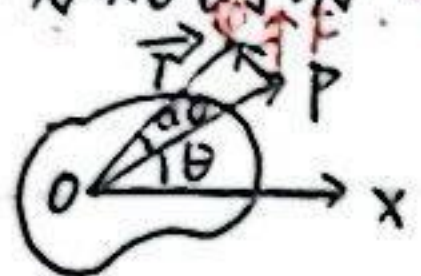
$$\therefore N = N_n = \frac{5 m g}{2}$$

三. 转动中的功和能

(1) 刚体的转动动能 $E_k = \frac{1}{2} J \omega^2$

$$\Delta m_i: \Delta E_k = \frac{1}{2} \Delta m_i v_i^2$$

$$\text{所有质元: } E_k = \frac{1}{2} \sum \Delta m_i v_i^2 = \frac{1}{2} \sum \Delta m_i (r_i \omega)^2 = \frac{1}{2} \left(\sum \Delta m_i r_i^2 \right) \omega^2 = \frac{1}{2} J \omega^2$$

(2) 力矩的功 $dA = T d\theta$ ($\vec{T} = J \beta \vec{\omega}$)

$$dA = \vec{F} \cdot d\vec{r} = F \cos \varphi |d\vec{r}| = F \cos \varphi (r d\theta)$$

$$\because F \cos \varphi \perp r \Rightarrow (F \cos \varphi) r = \vec{M}(\vec{r}) \text{ (力矩)}$$

$$dA = \vec{M} d\theta \quad (\text{力矩对物体作的功} = \text{力矩和角位移的乘积})$$

$$A = \int dA = \int_{\theta_1}^{\theta_2} T d\theta = \int_{\theta_1}^{\theta_2} J \beta d\theta = \int_{\omega_1}^{\omega_2} J \frac{d\omega}{dt} d\theta = \int_{\omega_1}^{\omega_2} J \omega d\omega$$

$$\Rightarrow A = \frac{1}{2} J \omega_2^2 - \frac{1}{2} J \omega_1^2 = E_{k2} - E_{k1} \quad \text{定轴转动的动能定理}$$

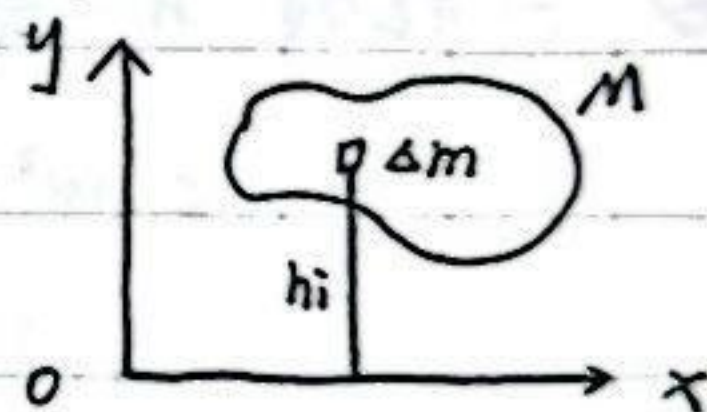
合外力矩做功 = 刚体转动动能的增量

(3) 刚体重力势能

$$E_p = \sum m_i g h_i = g \left(\sum \Delta m_i h_i \right) \frac{M}{M} = M g h_c$$

$$\text{刚体重力势能 } E_p = M g h_c \quad (\text{质心的高度})$$

它的全部质量都集中在质心时所具有的势能



(4) 机械能守恒

含刚体的系统. 只有保守内力做功 则系统机械能守恒

四. 刚体角动量定理 & 守恒定律

1. 刚体对定轴角动量: 一个质元: $L_{iz} = r_i \Delta m_i v_i = r_i^2 \Delta m_i \omega$

$$\vec{L} = J \vec{\omega}$$

$$\therefore L_z = \left(\sum \Delta m_i r_i^2 \right) \omega = J_z \omega$$



刚体对固定转动轴的角动量 = 该轴转动惯量 J 与角速度 omega 的乘积

2. 刚体绕定轴的 $\vec{L}$: $\vec{\tau}_i = \frac{d\vec{L}_i}{dt}$ $\vec{\tau}_z = \frac{dL_z}{dt}$

$$\text{质点系: } \sum \vec{\tau}_i = \sum \vec{\tau}_{i\text{外}} + \sum \vec{\tau}_{i\text{内}} = 0 \quad \vec{\tau} = \sum \vec{\tau}_{i\text{外}}$$

$$\tau_z = \sum \frac{dL_{iz}}{dt} = \frac{dL_z}{dt} = \frac{d}{dt} (J_z \omega)$$

$$\text{角动量定理: } \tau_z = \frac{dL_z}{dt}$$

(力矩) $\downarrow$

$$J_z \text{ 为常量: } \tau_z = J_z \beta$$

$$J_z \neq \text{常量: } \tau_z = \frac{d(J_z \omega)}{dt}$$

$$\text{冲量矩定理: } \tau_z dt = dL_z$$

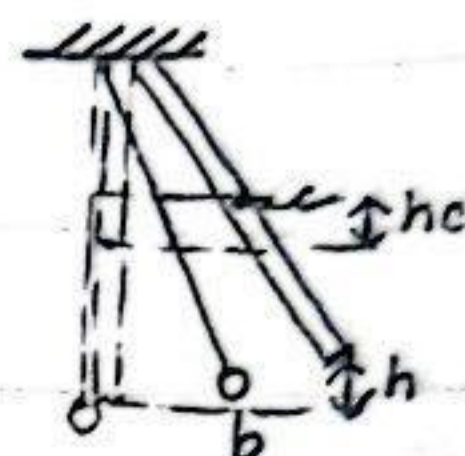
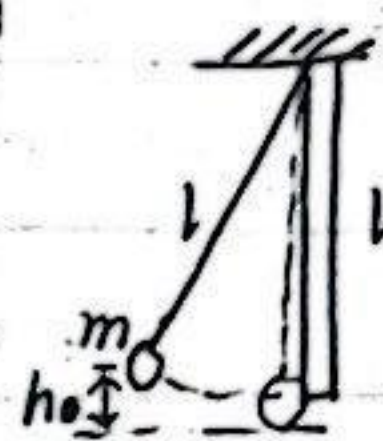
力矩冲量 (冲量矩) = 角动量

例4. 如图所示, 将单摆和一等长的匀质直杆悬挂在同一点. 杆的质量 m 与单摆的摆锤相同. 开始时直杆自然下垂, 将单摆的摆锤拉到高度 h_0 . 令它自静止状态下落, 于铅垂位置和直杆作弹性碰撞. 求碰后直杆下端达到高度 h .

三个阶段: ① 小球下落碰撞杆

② 碰撞过程 角动量守恒

③ 斜向上摆动



解: ① $v_0 = \sqrt{2gh_0}$

② 碰后角速度为 ω , 摆锤速度为 v'

$$\therefore ml(v_0 - v') = J\omega$$

$$\text{且机械能守恒 } \frac{1}{2}m(v_0^2 - v'^2) = \frac{1}{2}J\omega^2$$

$$\left. \begin{aligned} & \\ & \end{aligned} \right\} \Rightarrow \begin{aligned} v' &= \frac{v_0}{2} \\ \omega &= \frac{3v_0}{2l} \end{aligned}$$

③ $\therefore$ 摆锤 $h' = \frac{h_0}{4}$

$$\text{杆: } \frac{1}{2}J\omega^2 = mgh_c \Rightarrow h = 2h_c = \frac{3h_0}{2}$$

例5. 一转动惯量为 I_1 的轮绕固定轴靠惯性以角速度转动, 另一转动惯量为 I_2 且处于静止状态的轮突然与前者合并绕同一轴以相同 ω 转动. 求共同转动 ω .

$\therefore$ 不受外力矩, 角动量守恒

$$I_1 \omega = (I_1 + I_2) \omega'$$

$$\omega' = \frac{I_1 \omega}{I_1 + I_2}$$

$$\tau dt = dL \Rightarrow \int_0^t \tau dt = \int dL = L_t - L_0 \in L = J\omega$$

$$\Rightarrow \int_0^t \tau dt = \int_{L_0}^{L_t} dL = L_t - L_0 = J\omega - J_0\omega_0$$

$$\int_0^t \tau dt = J\omega - J_0\omega_0$$

动量定理 $\int \vec{F} dt = \int d\vec{p} = \vec{p}_t - \vec{p}_0 = m\vec{v}_t - m\vec{v}_0$

3. 角动量守恒定律 用动量守恒 $\checkmark$ 角动量一定不守恒

当合外力矩 $\tau = 0$ 则: $L_t = L_0 = \text{常量} \rightarrow$ 角动量守恒

(1) J 为常量, L 为常量 则 ω 不变

(2) L 为常量, $J\omega = \text{常量}$ $J_1\omega_1 = J_2\omega_2$ 可知 $J \uparrow \omega \downarrow$ 或 $J \downarrow \omega \uparrow$

(3) 对多个刚体组成的系统也成立 同一转轴

初时刻 $L_0 = 0$

t 时刻 $L_t = 0$ $\left\{ \begin{array}{l} \text{系统无转动 } L_1 = L_2 = 0 \\ \text{有 } L_1 + L_2 = 0 \Rightarrow L_1 = -L_2 \end{array} \right.$

$$\Rightarrow L_1 = -L_2 \quad (J\omega_1 = -J_2\omega_2)$$

例: 一质量 m 子弹水平射入一静止长棒, 穿出后 v 损失 $\frac{3}{4}$, 求子弹穿出棒后角速度 ω .

不轻易使用动量定理

法一:

解: 阻力冲量 $\int f dt = m(v - v_0) = -\frac{3}{4}mv_0$

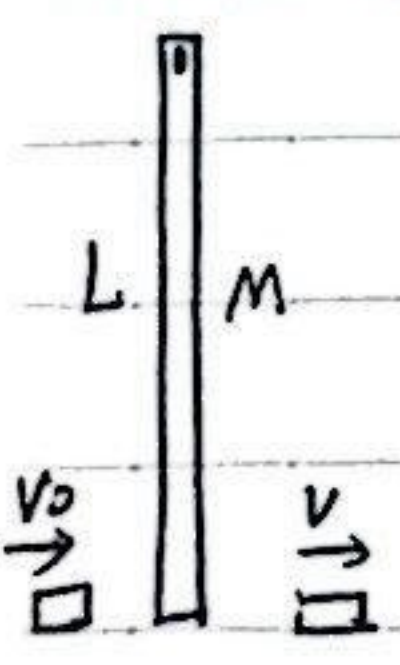
总动量不守恒!

阻力矩冲量 $L \cdot \int f' dt = J\omega$

转轴上有支持力的作用

$$f' = -f \Rightarrow \omega = \frac{3mv_0L}{4J} = \frac{9mV_0}{4ML}$$

角动量守恒



法二: 角动量守恒

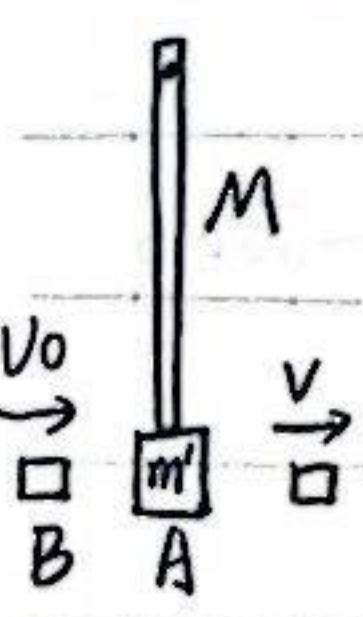
$$mv_0L = mVL + J\omega \Rightarrow \omega = \frac{9mV_0}{4ML}$$

动量守恒 (杆子重量集中在下端 / 轻绳) $\Rightarrow$ 不考虑转动效应

若 $m = 0$, 则 $J = 0$

A 获得动能使杆一起转动时由 $\tau = J\beta \therefore$ 无力矩 / 水平力

$\therefore$ A, B 系统 动量, 角动量都守恒

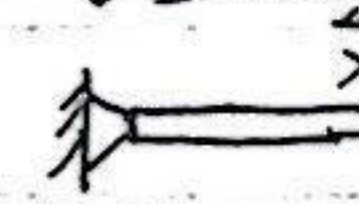


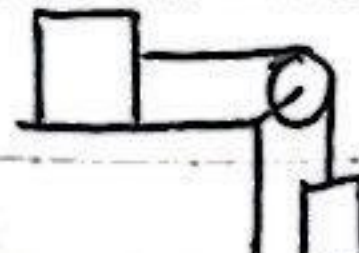
第三章小结

刚体的运动 { 平动
转动 **本章主要讨论** 一定轴转动 θ, ω, β
质心位矢 $\vec{r}_c = \frac{\sum m_i \vec{r}_i}{M} (x, y, z)$

刚体的定轴转动 { 刚体定轴转动定律 { $\vec{\tau} = \vec{r} \times \vec{F}$ 力矩
定律证明
 $\vec{\tau} = J\beta, J = \sum m_i r_i^2$

转动惯量的计算 J { 质点 MR^2 、圆环 MR^2 、圆盘 $\frac{1}{2}MR^2$
杆 $\frac{1}{3}MR^2$ (一端) $\frac{1}{12}MR^2$ (质心)
球体 $\frac{2}{5}MR^2$ 球壳 $\frac{2}{3}MR^2$
平行轴定理 $J_A = J_c + md^2$
垂直轴定理 $J_z = J_x + J_y$

定轴转动定律应用 { 

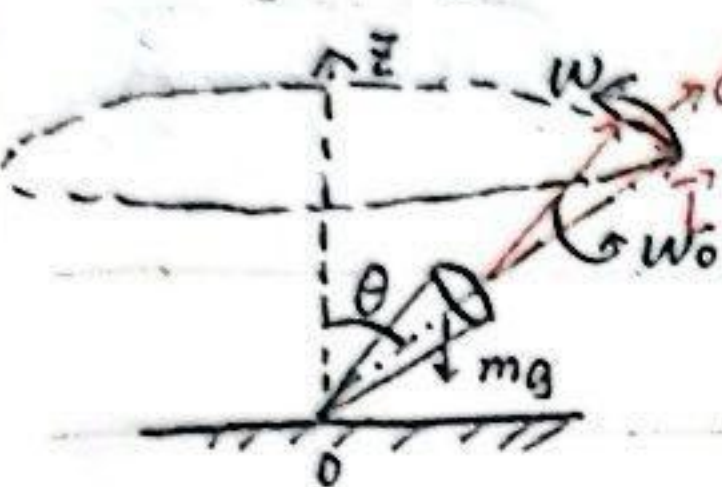



转动中的功与能 { 转动动能 $E_k = \frac{1}{2} J \omega^2$
力矩的功 $dA = F \cos \varphi (r d\theta) = \vec{\tau} d\theta$
动能定理 $A = \frac{1}{2} J \omega_2^2 - \frac{1}{2} J \omega_1^2 = E_{k2} - E_{k1}$
刚体重力势能 $E_p = Mgh_c$

角动量定理与角动量守恒定律 { 角动量 $\vec{L} = \vec{r} \times \vec{p} = m \vec{v} \times \vec{r}$ (质点对定点)
 $\vec{L} = J\omega$ (刚体对定轴)
角动量定理 $\vec{\tau} = \frac{d\vec{L}}{dt}, \vec{\tau} dt = d\vec{L}$
冲量矩 $\int_0^t \vec{\tau} dt = \vec{L}_t - \vec{L}_0 = J\omega - J\omega_0$
角动量守恒定律 $\vec{\tau} = 0$ 时 $\vec{L}_t = \vec{L}_0 = \text{常量}$

进动 { ① 自身旋转并绕定轴转动 $d\vec{L}$ 使 $\vec{L}$ 改变方向
② 应用 $\omega = \frac{\tau}{J\omega \sin \theta}$

进动



1. 进动: 陀螺在绕本身对称轴线转动的同时, 对称轴还将绕竖直轴 Oz 转动, 这种回转现象称为进动

2. 产生原因: 重力对 O 点力矩为 $\vec{\tau}$, 方向 $\vec{r} \times m\vec{g} \Rightarrow \otimes$

$$\because \vec{\tau} \perp \vec{L} \therefore d\vec{L} \perp \vec{L} \quad (d\vec{L} \text{ 使 } \vec{L} \text{ 改变方向})$$

$$\therefore \text{自转轴改变方向 } d\vec{L} = \vec{L} \cdot d\alpha \Rightarrow \frac{d\vec{L}}{dt} = L \cdot \frac{d\alpha}{dt} \Rightarrow \vec{\tau} = L\omega$$

$$\therefore \omega = \frac{\vec{\tau}}{L} = \frac{\vec{\tau}}{J\omega_0 \sin\theta}$$

△ 进动的应用: (1) 飞机自动驾驶、轮船稳定器

(2) 自旋电子在外磁场中的进动 (抗磁性)

(3) 子弹、炮弹、导弹在飞行时绕自身的轴旋转.

遇到阻力, 偏离轴向后产生进动, 总的运动仍保持原方向前进